

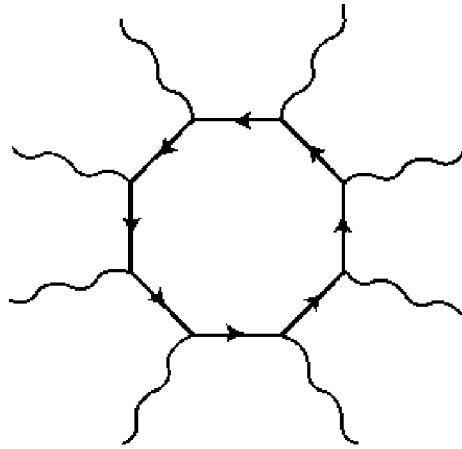


Figure 6.3. An eighth-order graph for the vacuum-to-vacuum amplitude with particles interacting only with an external field. In this diagram the external field is represented by wiggly lines. There are $7!$ such diagrams, differing only by relabelling the vertices, and not counting as different those labellings that simply rotate the ring. The factor $1/8!$ from the Dyson formula (6.1.1) is therefore not entirely cancelled here, leaving us with an extra factor $1/8$.

pairings

$$[a(2')\psi^\dagger(x)] [a(1')\psi^\dagger(y)] [\psi(y)a^\dagger(1)] [\psi(x)a^\dagger(2)] \quad (6.1.20)$$

and

$$[a(1')\psi^\dagger(x)] [a(2')\psi^\dagger(y)] [\psi(y)a^\dagger(1)] [\psi(x)a^\dagger(2)] . \quad (6.1.21)$$

(See Figure 6.4.) To go from (6.1.19) to (6.1.20) requires an *even* permutation of fermionic operators. (For instance, move $\psi(x)$ past three operators to the right, and then move $a(1')$ past one operator to the right.) Thus there is no extra minus sign in the contribution of the pairing (6.1.20). This in itself is not so important; the *overall* sign of the S -matrix does not matter in transition rates, and in any case depends on sign conventions for the initial and final states. What is important is that the contributions of pairings (6.1.20) and (6.1.21) have opposite sign, as can be seen most easily by noting that the only difference between these two pairings is the interchange of two fermionic operators, $a(1')$ and $a(2')$. In fact, this relative minus sign is just what is required by Fermi statistics: it makes the scattering amplitude antisymmetric under the interchange of particles $1'$ and $2'$ (or 1 and 2).

However, it must not be thought that all sign factors can be related in such a simple way to the antisymmetry of the final or initial states, even in the lowest order of perturbation theory. To illustrate this point, let's now consider fermion–antifermion scattering, $12^c \rightarrow 1'2'^c$, to second order in the same interaction (6.1.18). The fermionic operators in the second-order